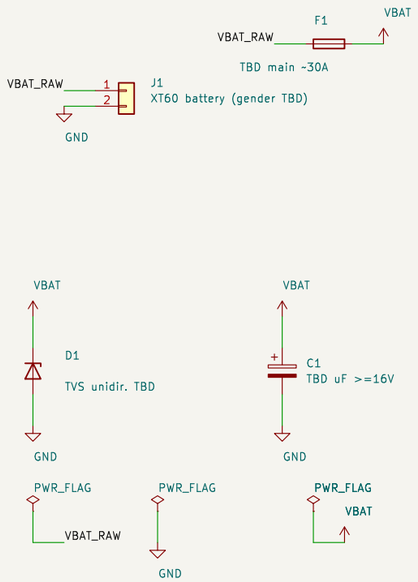
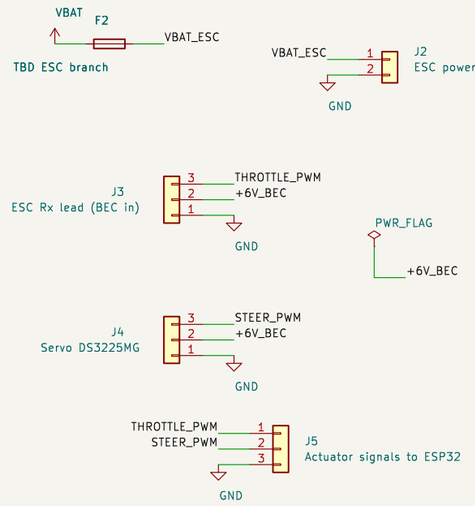
A Battery input (2S LiPo, 6.0–8.4 V)

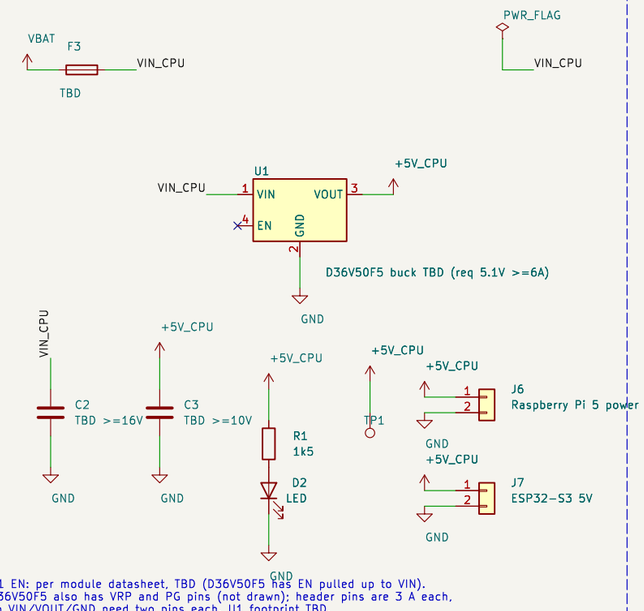


B ESC and servo (servo power = ESC BEC, 6 V / 3 A)



+6V_BEC comes only from the ESC BEC.
Never connect it to any 5 V rail.
J5: THROTTLE_PWM = GPIO5, STEER_PWM = GPIO4 (3.3 V logic).

C Compute 5 V (+5V_CPU: Pi 5 + ESP32 only)

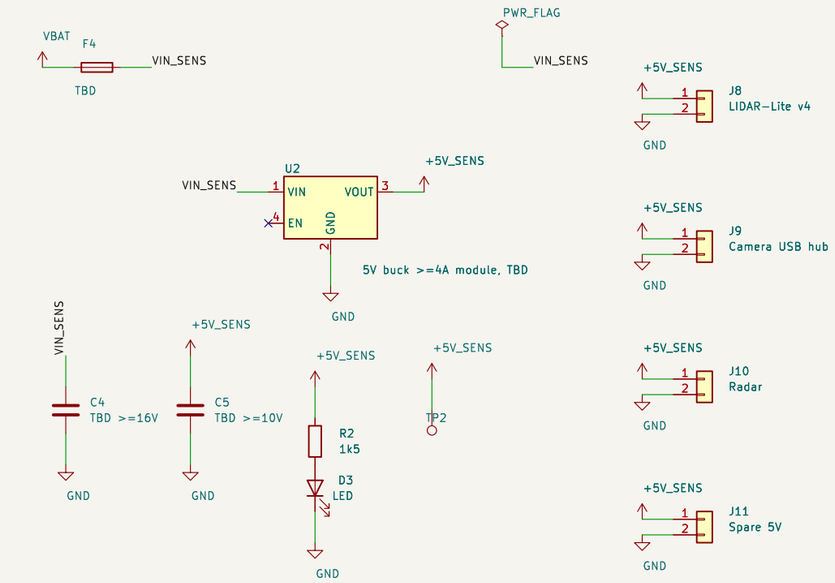


U1 EN: per module datasheet, TBD (D36V50F5 has EN pulled up to VIN).
D36V50F5 also has VRP and PG pins (not drawn): header pins are 3 A each,
so VIN/VOUT/GND need two pins each. U1 footprint TBD.
Plan asks 5.1 V, >=6 A; D36V50F5 is 5.0 V +/-4 %, 5.5 A typ (open Q4).
Reverse polarity protection TBD (P-FET or ideal diode, plan step 7);
D36V50F5 itself has input reverse protection to 40 V (open Q6).

Notes

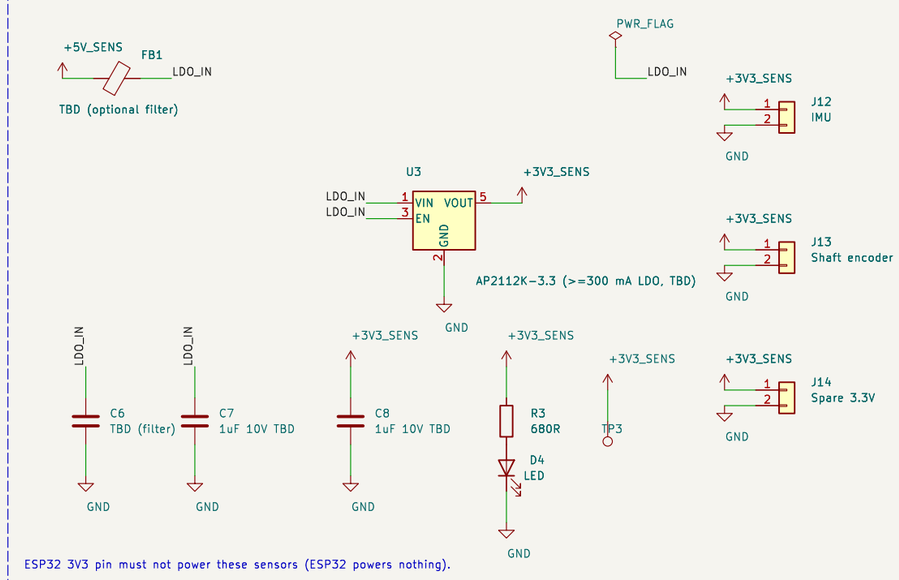
1. Battery: 2S LiPo, 6.0–8.4 V (ESC LVC cuts at 6.0 V), XT60.
2. All fuses and protection values TBD;
   positions are proposals.
3. Single common GND; star point at J1 in layout.
4. +6V_BEC is ESC-supplied, isolated from 5 V rails.
5. Signals are 3.3 V logic or lower.
6. Rails are project power symbols (rc_power lib):
   pin name = Value = net name (KiCad 7–10 safe).
7. PWR_FLAG on VBAT_RAW, GND, VBAT, VIN_CPU,
   VIN_SENS, LDO_IN, +6V_BEC.
8. Net classes (kicad_pro): HighCurrent = *VBAT*, GND;
   Power5A = +5V_CPU, *VIN_CPU;
   Power3A = +5V_SENS, *VIN_SENS, *+6V_BEC
   ('*' covers the '/' prefix KiCad gives local labels).
   20 A path: zones or off-board wire (open Q1).
9. Spec: docs/schematics/power_board_plan.md.
   Values marked TBD are unconfirmed.

D Sensor 5 V (+5V_SENS: heavy sensors + 3.3 V LDO)



U2 EN: per module datasheet, TBD. U2 footprint TBD (module pinout).

E Sensor 3.3 V (+3V3_SENS: LDO from +5V_SENS)



ESP32 3V3 pin must not power these sensors (ESP32 powers nothing).



Baseline generated programmatically; project KiCad version 10.0
Spec: docs/schematics/power_board_plan.md
Draft for review, values marked TBD are unconfirmed
gvo27/racecar

Sheet: /
File: rc_power_board.kicad_sch

Title: RC Car Power Distribution Board

Size: A3	Date: 2026-10-28	**Rev: A (draft)**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1